

[1128A62]

SARDAR PATEL UNIVERSITY**BCA Sem-IV EXAMINATION, 2017****2.00 pm to 5.00 pm****COMPUTER BASED NUMERICAL & STATISTICAL METHOD (US04FBCA01)****Date: 3/04/2017****Maximum Marks: 70****Q-1 Write the correct option in the answer book.****[10]**

- (1) If $f(a) < 0$, $f(b) > 0$ and if $x_0 \in (a, b)$ is first approximation with $f(x_0) < 0$ then in bisection method _____
(a) x_0 is to be replaced by a (b) a is to be replaced by c
(c) b is to be replaced by x_0 (d) x_0 is to be replaced by b
- (2) The number 0.01850×10^3 has _____ significant digits
(a) 3 (b) 4 (c) 5 (d) 6
- (3) All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a _____.
(a) Polynomial (b) Equation (c) Algorithm (d) None of the above
- (4) _____ Method is used if the estimated value lies towards the end of the difference table.
(a) Divided difference (b) Forward difference
(c) Backward difference (d) None of the above
- (5) y depends on x can be written as _____
(a) $f(x)$ or y_x (b) $f(xy)$ (c) $f(yx)$ (d) none of the above
- (6) _____ is called the backward difference operator.
(a) Δ (b) ∇ (c) $\emptyset$ (d) $\cup$
- (7) We can find solution of system of linear, algebraic equations using _____.
(a) Newton-Raphson method (b) Bisection method
(c) Gauss-Seidel method (d) None of these
- (8) The system of linear equation $AX = B$ can be solved by matrix inversion method only if _____.
(a) $A \neq 0$ (b) $|A| \neq 0$ (c) $|A| = 0$ (d) A is symmetric
- (9) Forecasts _____.
(a) become more accurate with longer time horizons
(b) are rarely perfect
(c) are more accurate for individual items than for groups of items
(d) all of the above
- (10) Gradual, long-term movement in time-series data is called _____.
(a) seasonal variation (b) cycles
(c) trends (d) exponential variation

Q:2 Answer the following in short. (Any Ten)

[20]

- (1) Find the next iterative value of the root of using secant method, if the initial guesses are 3 and 4
- (2) Define Relative error and absolute error
- (3) Describe the stopping rules to obtain approximate solution for given non-linear equations.
- (4) Use the secant method to obtain approximate solution of the equation $X^3 - 5x - 3 = 0$. [initial approx. 2 & 3]
- (5) Define Interpolation.
- (6) Explain Divided difference table.
- (7) List the component of Time series.
- (8) Solve the system of equation.
$$\begin{aligned} 2x + 3y &= -10 \\ -x + 4y &= -4 \end{aligned}$$
- (9) If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- (10) What is Time Series?
- (11) What do you mean by Secular trend?
- (12) What do you mean by random or irregular Variation?

Q-3

- (a) Find the root of equation $x^3 - 4x - 9 = 0$ correct up to four decimal places using Bisection method. **[05]**
- (b) Find the root of equation $x^3 - x^2 - 1 = 0$ correct up to four decimal places using Iterative method. (Take $a=1.5$). **[05]**

OR

Q-3

- (c) Find the root of equation $x^3 - x - 1 = 0$ correct up to four decimal places using False Position method. **[05]**
- (d) Find the root of equation $xe^x - 1 = 0$ correct up to four decimal places using Secant method. **[05]**

Q-4

- (a) Using Lagrange's method for interpolation find y for $x=2$ [05]

X	1	3	4	6
Y(x)	-3	9	30	130

- (b) Using Newton's forward formula find $y(2.05)$ [05]

x	2.0	2.1	2.2	2.3
y(x)	11.0	12.261	13.648	15.167

Q-4

OR

- (c) The following table gives the census population of a town for the years 1931 to 1971. Estimate the population for the year 1935 and year 1965 by using an appropriate interpolation formula. [10]

Year	1931	1941	1951	1961	1971
Population	46	66	81	93	101

Q-5

- (a) Solve the following system of equations using Gauss-Seidel method. [05]

$$10x_1 + x_2 + 2x_3 = 44$$

$$2x_1 + 10x_2 + x_3 = 51$$

$$x_1 + 2x_2 + 10x_3 = 61$$

- (b) Explain the Matrix Inversion method for solution of system of linear equations. [05]

Q-5

OR

- (c) The distance (s) covered by a car in given time (t) is given in the following table: [05]

Time(minutes)	10	12	14	16	18
Distance(km)	12	15	20	27	37

Determine the speed of the car at $t=13$ minutes.

- (d) Use Backward difference formula to compute $\frac{dy}{dx}$ at $x=7$ and $x=7.5$, [05]

x	3	4	5	6	7	8
y	4	6.6	7.7	9.0	10.5	12.2

From the following table:

- Q.6** Obtain seasonal indices using simple average method.
(a)

[06]

Year	Q-I	Q-II	Q-III	Q-IV
1990	30	81	62	119
1991	33	104	86	171
1992	42	153	99	221
1993	56	172	129	235

- (b)** Obtain the trend from the time series given below by method of moving average of 4 years **[04]**

Year	1958	1959	1960	1961	1962	1963	1964	1965	1966	1967
Y	50.0	36.5	43.0	44.5	38.9	38.1	32.6	41.7	41.1	33.8

OR

- Q.6** Obtain seasonal indices using ratio to moving average.
(c)

[06]

Year	Q-I	Q-II	Q-III	Q-IV
1990	30	40	36	34
1991	34	52	50	44
1992	40	58	54	48
1993	54	76	68	62

- (d)** What is time series? List various components of time series and explain one of them. **[04]**

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